Ex:13 Network Address Translation (NAT)

AIM:

To configure Network Address Translation (NAT) on a router to allow devices with private IP addresses to access external resources using a public IP.

ALGORITHM:

1. **Set Up the Network**:
   - Drag and drop a **Router**, **Switch**, and **PCs** into the workspace.
   - Connect the router to both the internal network (via a switch) and the external network (using a server to simulate the Internet).

2. **Configure Router Interfaces**:
   - Assign IP addresses to the router's internal and external interfaces:
     - **Internal (Private IP)**: `192.168.1.1/24`
     - **External (Public IP)**: `209.165.200.1/24`

     Example:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip address 209.165.200.1 255.255.255.0
Router(config-if)# no shutdown
```

3. **Define Access List for NAT**:
   - Create an access list to permit the private IP range to be translated by NAT.

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

4. **Enable NAT on the Router**:
   - Configure NAT on the router, mapping the private IP range to the public IP interface.

```
Router(config)# ip nat inside source list 1 interface
gigabitEthernet0/1 overload
```

5. **Configure Inside and Outside Interfaces**:

Mark the router's internal interface as **inside** and the external interface as **outside**.
Example:

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip nat inside

Router(config)# interface gigabitEthernet0/1
```

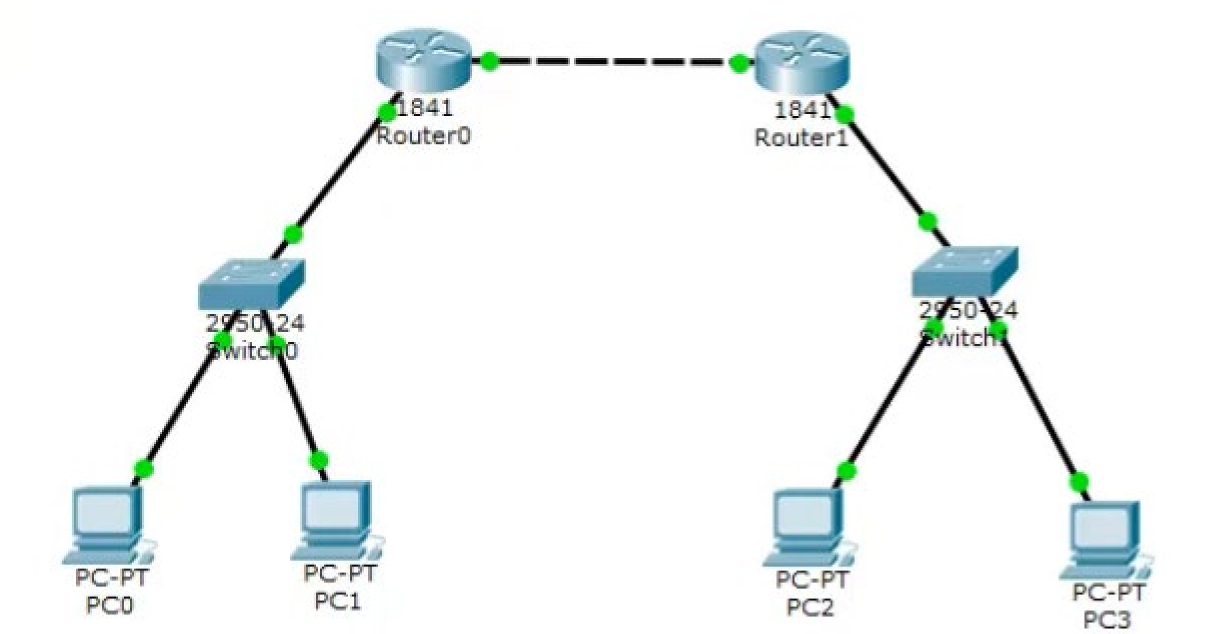
```
Router(config-if)# ip nat outside
```

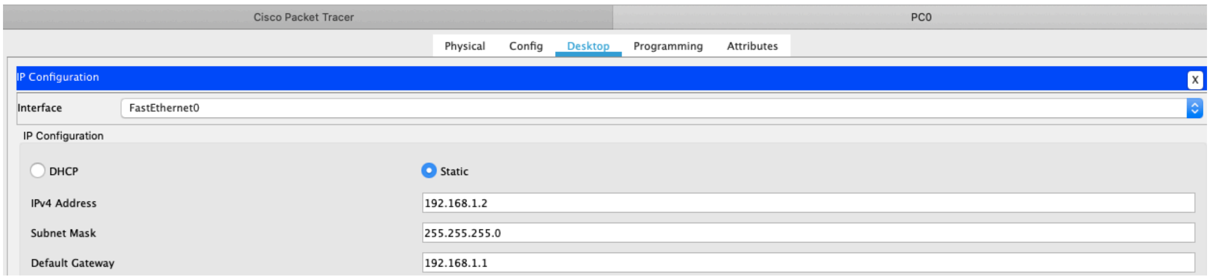
6. **Test the NAT Configuration**:
   - From a PC in the internal network, attempt to access the server (Internet).
   - Use **ping** or open a web page to test connectivity.

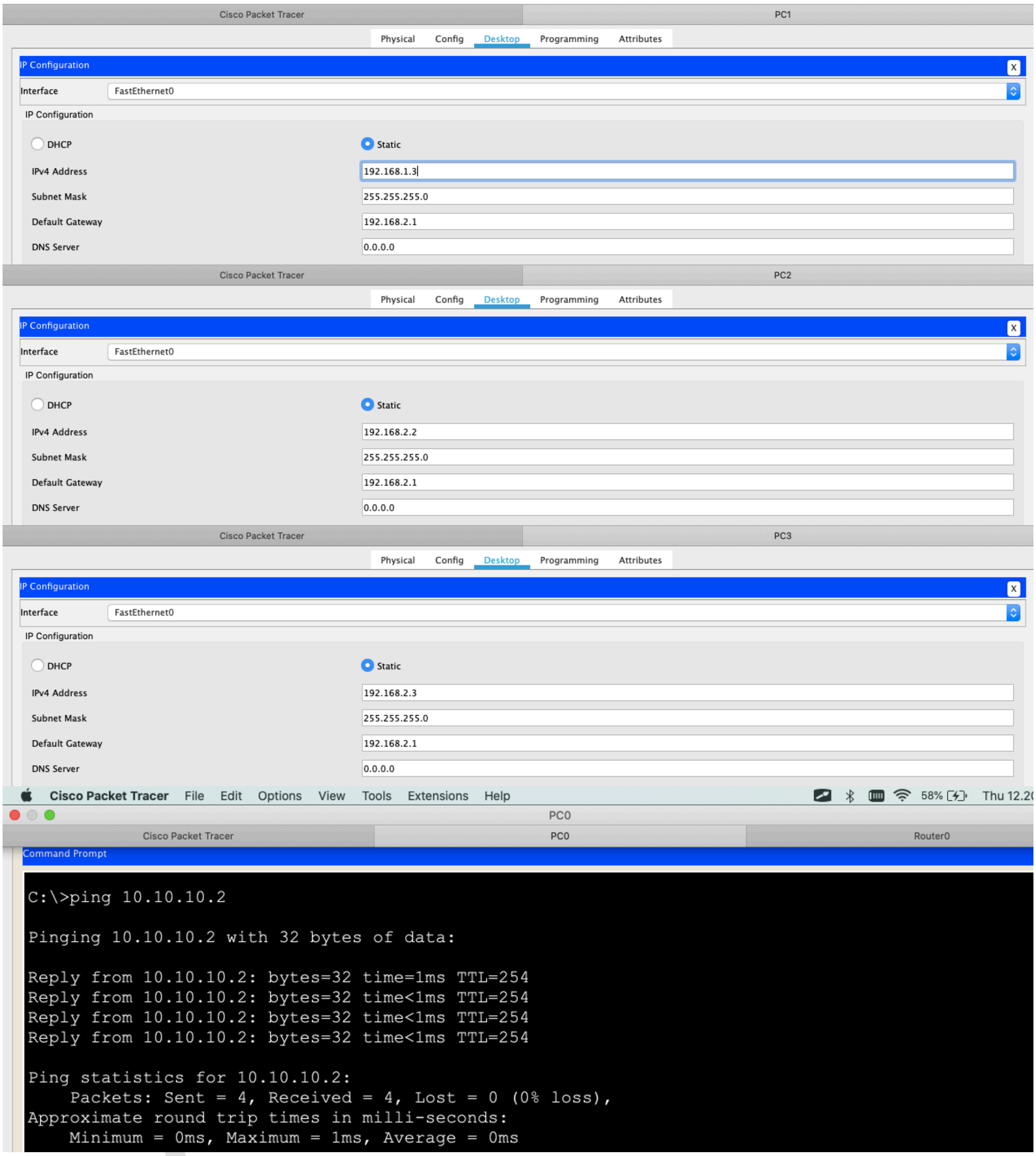
7. **Verify NAT Translations**:
   - Check the NAT translations on the router:

```
Router# show ip nat translations
```

OUTPUT:







RESULT:

- The NAT configuration successfully allows devices with private IP addresses in the internal network to communicate with the external network using the router's public IP.
- The **show ip nat translations** command shows the active NAT translations between private and public IP addresses.
- Successful **ping** responses or web access from internal PCs to external servers confirm that NAT is working as expected.